

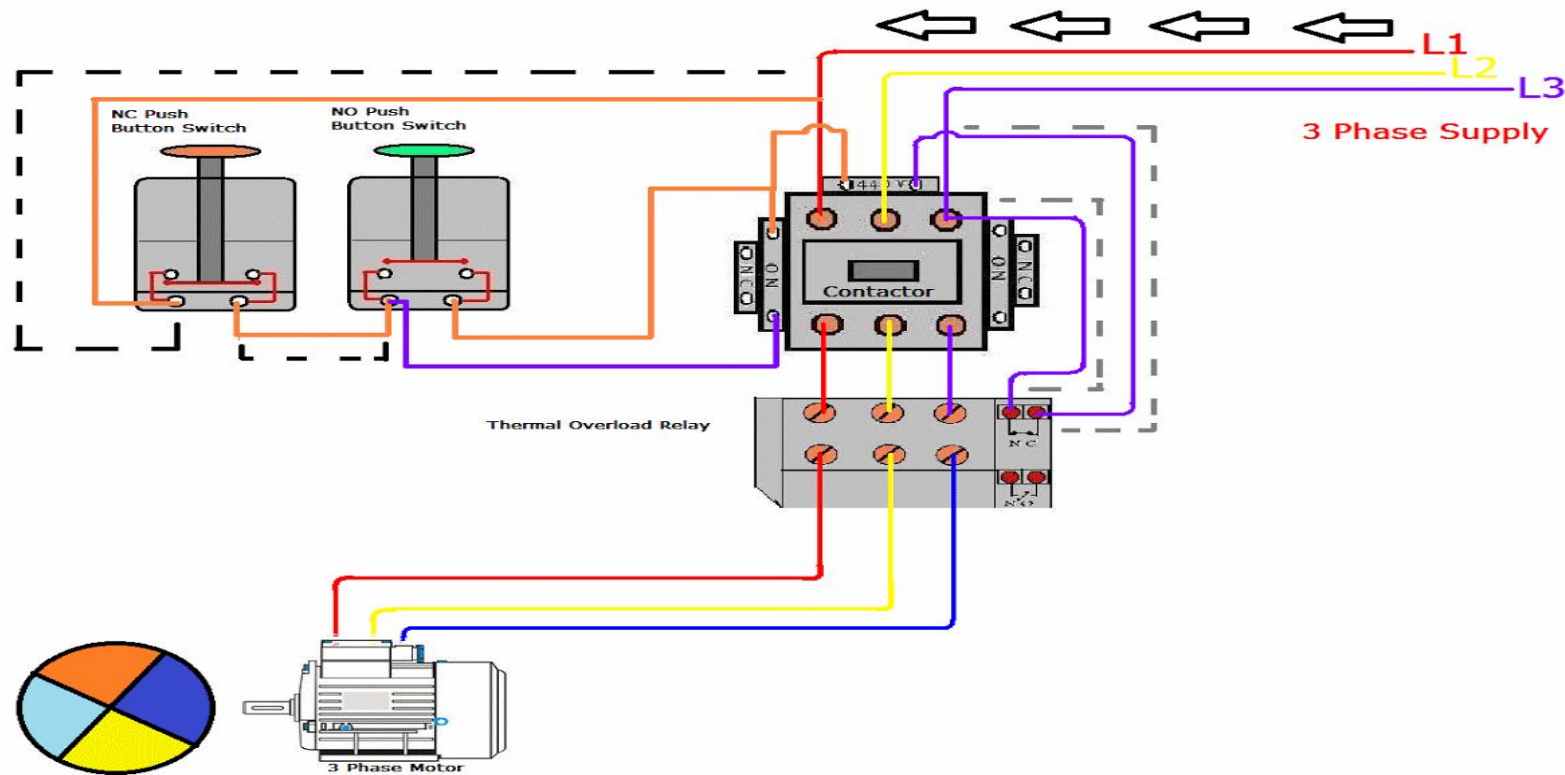


ELECTROMECHANICAL SYSTEM

CHAPTER 2: DEFINITION AND PRINCIPLE OF MOTOR
OPERATIONS

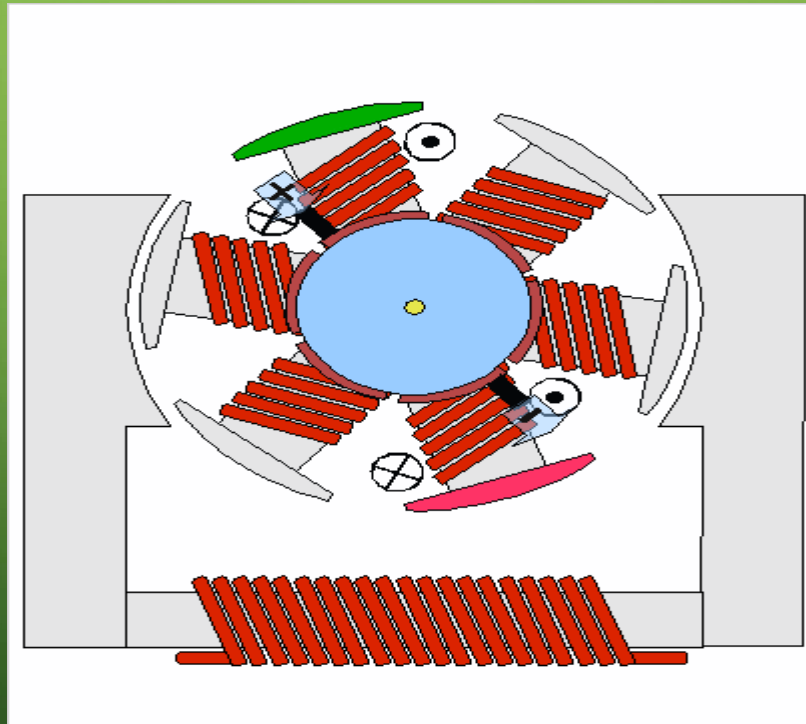
MOTOR CONTROL

A MOTOR CONTROL CIRCUIT CAN BE DEFINED AS A MEANS OF SUPPLYING POWER TO AND REMOVING POWER FROM A MOTOR.



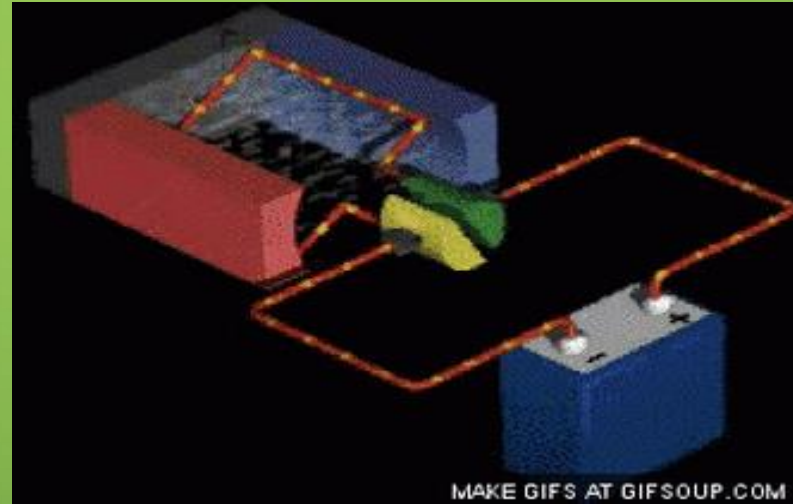
ELECTRIC MOTOR

It is a machine which converts electric energy into mechanical energy or work and requires electric energy input.

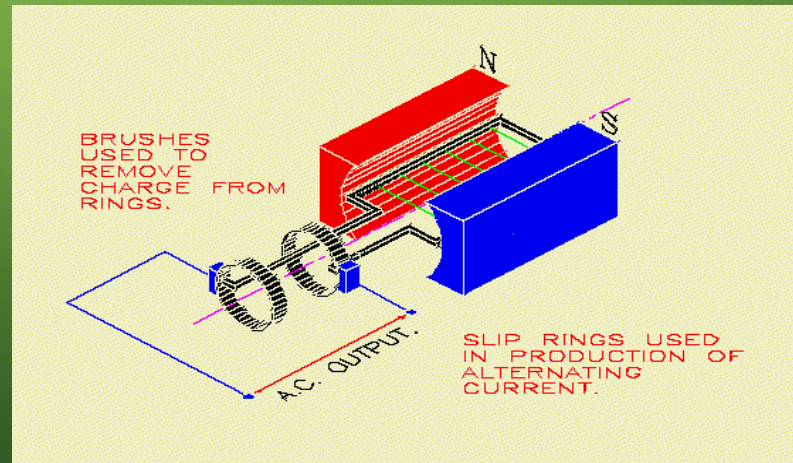


TWO TYPES OF ELECTRIC MOTORS

1. DC Motor



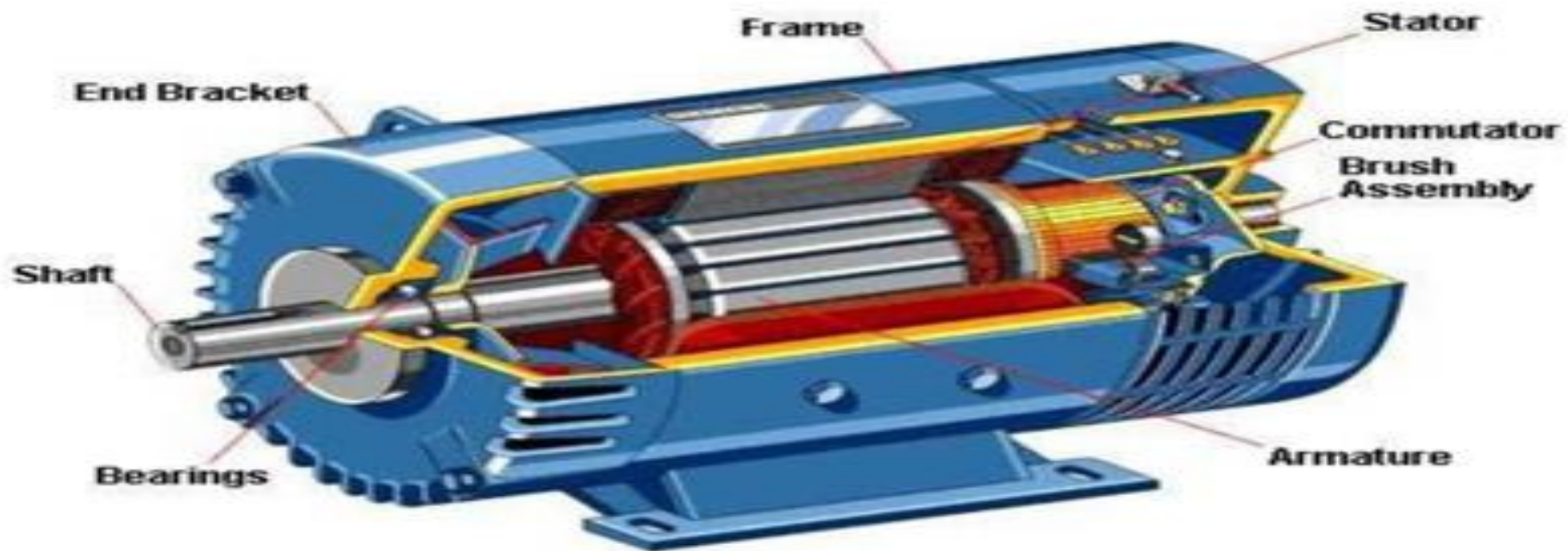
2. AC Motor



1. DC MOTORS

- Operate on Direct Current from a battery or DC power supply.
- They work on the principle that a conductor in a magnetic field experiences a mechanical force.

1. DC MOTORS



DC Motor Breakdown

1. DC MOTORS

3 Basic Parts

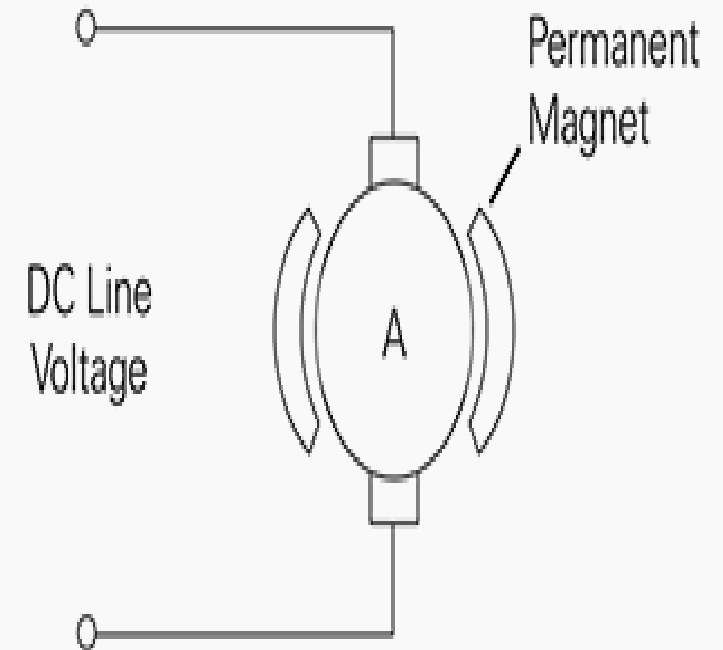
1. Armature – The rotating part with wire coils around a core and a shaft that rotates on bearings.
2. Stator – The stationary part containing coils of wire.
3. Commutator – Terminated points on the armature where brushes make contact to transfer current from stationary to rotating parts.

1. DC MOTORS

4 Basic types of DC motors

1. Permanent Magnet DC Motor — Uses a magnet for field flux.

- **Advantages:** Excellent starting torque and good speed regulation.
- **Disadvantages:** Limited load capacity, typically used in low horsepower applications; torque is limited to 150% of rated torque to avoid magnet demagnetization.

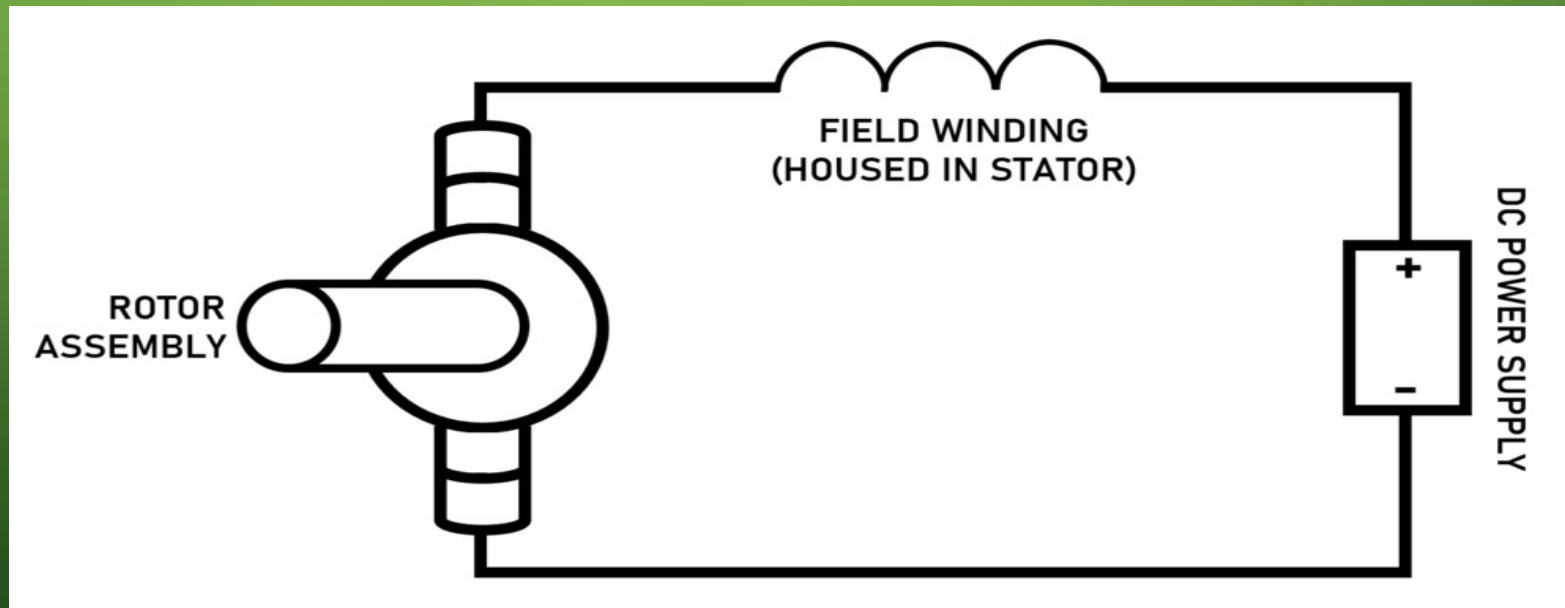


Permanent Magnet Motor

1. DC MOTORS

4 Basic types of DC motors

- 2. Series DC Motor — Provides high starting torque.
- **Advantages:** Capable of moving large shaft loads when first energized.
- **Feature:** Field winding and armature are connected in series

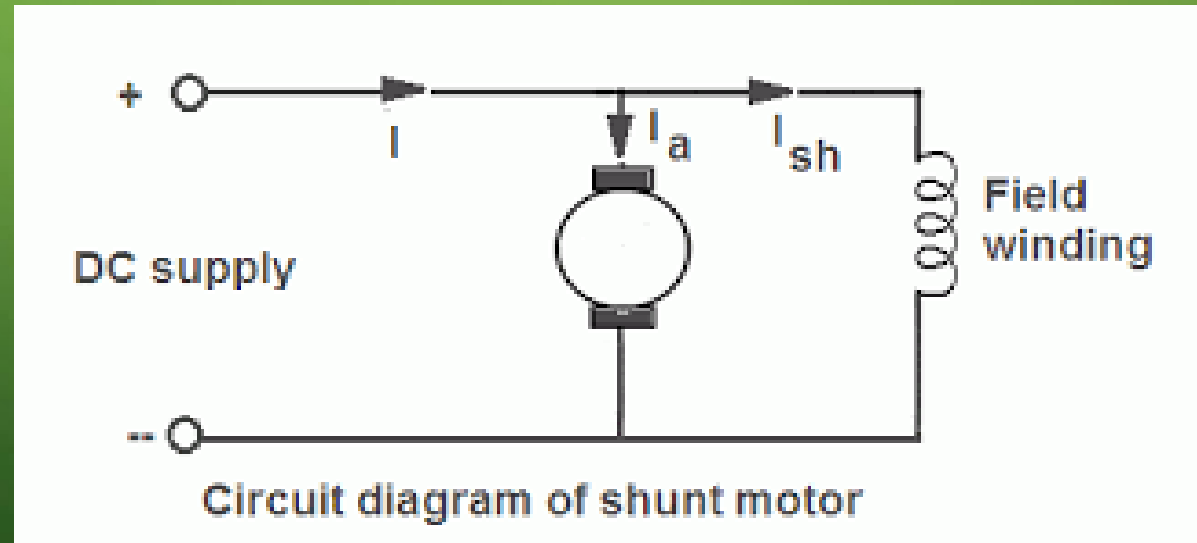


1. DC MOTORS

4 Basic types of DC motors

3. Shunt DC Motor – Field winding is connected in parallel with the armature.

- **Feature:** Field windings consist of many turns of wire.

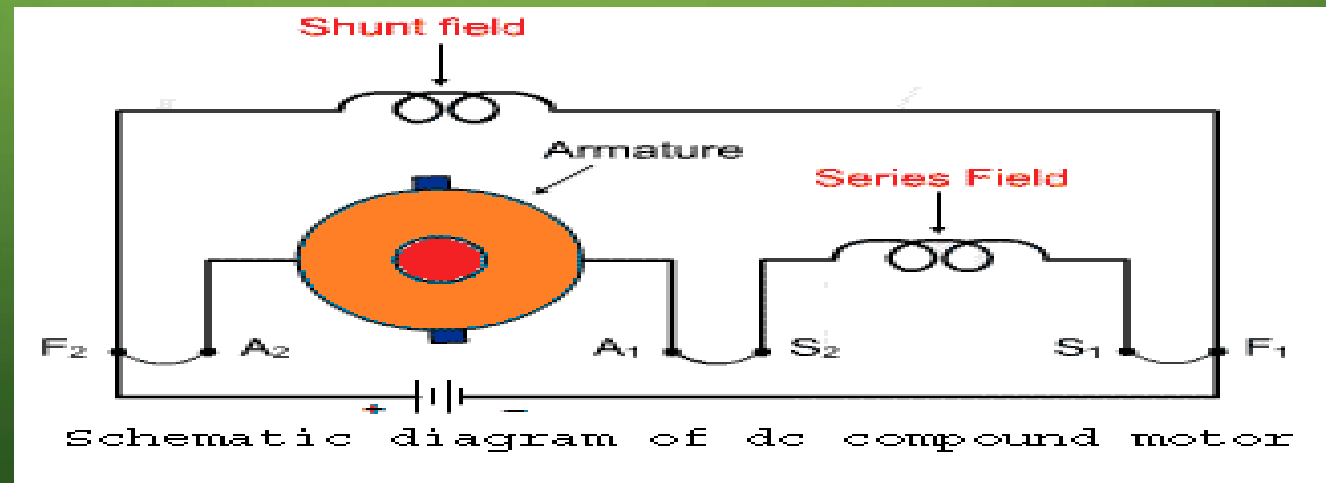


1. DC MOTORS

4 Basic types of DC motors

4. Compound DC Motor – Combines series and shunt motor features.

- **Series Field Winding:** Connected in series with the armature.
- **Shunt Field Winding:** Connected in parallel with the armature.
- **Benefits:** Combines high torque and regulated speed.



1. DC MOTORS

DC Motors for Mobile Robots

1. **Brushed DC motor** - Uses permanent magnets in the rotor and electromagnets in the stator.

- **Commutator:** Detects rotor orientation and flips the current using brushes.
- **Function:** Brushes rotate with the rotor to maintain current flow and motor operation.

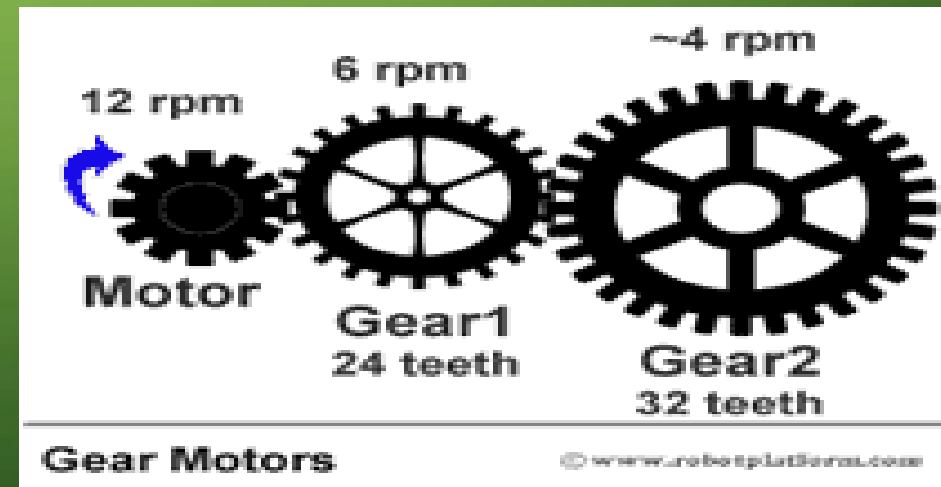


1. DC MOTORS

DC Motors for Mobile Robots

2. **Geared DC motor** - DC motor coupled with gears to increase torque while reducing speed.

- **Application:** Commonly used in robots to pull weight and components.
- **Gear Mechanism:** Motor shaft connects to gears, decreasing rpm and increasing torque.



1. DC MOTORS

DC Motors for Mobile Robots

3. **Brushless DC motors** - Rotor uses permanent magnet, stator uses electromagnet, and Hall Effect sensors detect rotor orientation.

- **Advantages:** Higher torque and greater speed than brushed motors, ideal for robots.
- **Disadvantages:** Expensive due to design complexity, requires a controller for speed and rotation.



1. DC MOTORS

DC Motors for Mobile Robots

4. Servo Motors - DC motors with feedback control circuitry, gear system, and position sensing (potentiometer).

- **Function:** Moves motor shaft to a desired position based on control signal (pulse).
- **Rotation:** Limited to a specific range (usually 200°); can be modified for continuous rotation.
- **Wiring:** Requires three wires: Ground, Power, and Control pulse.



1. DC MOTORS

DC Motors for Mobile Robots

5. **Stepper motors** - Brushless motors that divide rotation into discrete steps with electrical pulses.

- **Function:** Rotates in steps per revolution, jumping each step with a pulse.
- **Torque:** Similar to brushed DC motors but with less torque.
- **Feedback:** No complex position feedback required.



1. DC MOTORS

DC Motors for Mobile Robots

6. **Linear DC motor** - Stator unwrapped into a flat coil track.

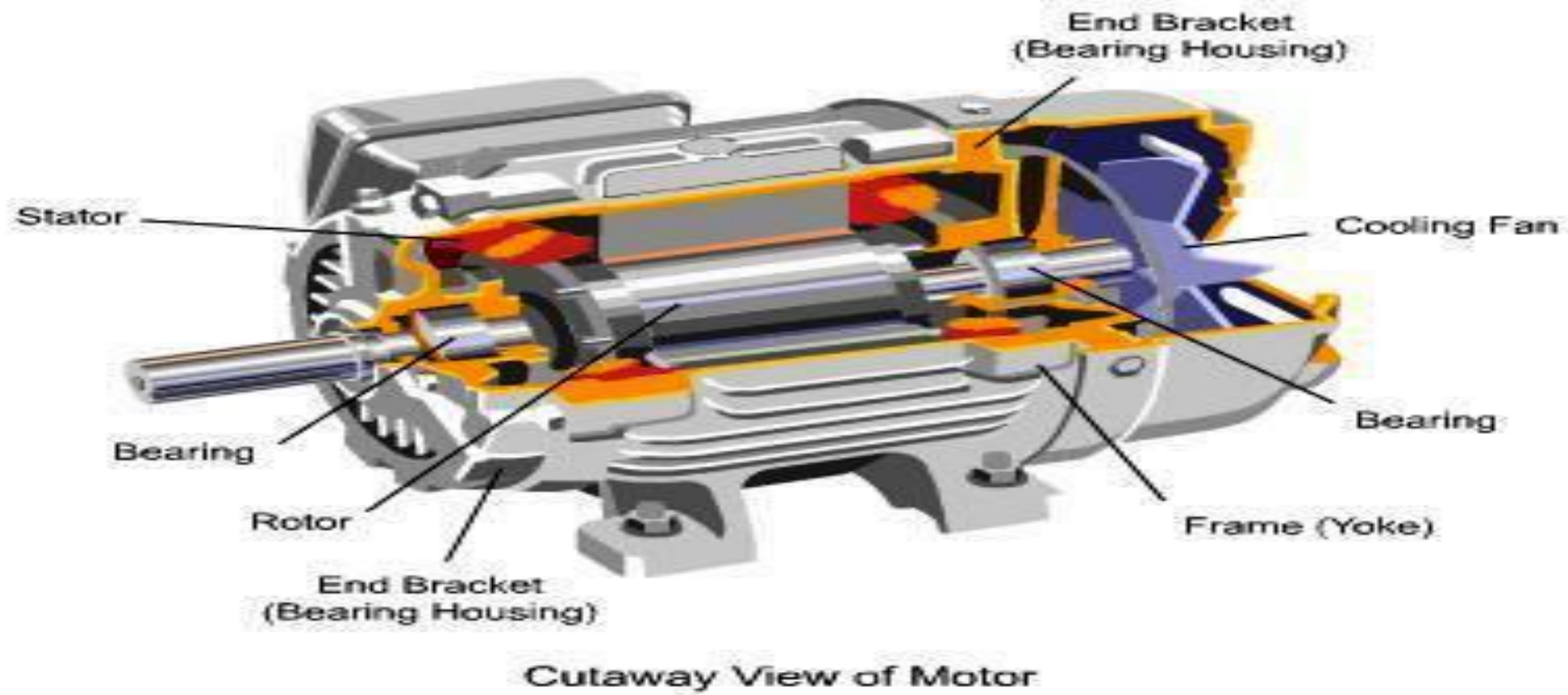
- **Function:** Rotor moves in a straight line over the stator.



2. AC MOTORS

- Convert AC electrical power into mechanical power.
- Operate on Alternating Current, oscillating between positive and negative.
- Principle: Interaction between the revolving magnetic field in the stator and the opposing magnetic field in the rotor.

2. AC MOTORS



2. AC MOTORS

Basic Parts

1. Stator – fixed outer portion
2. Rotor – spins inside the AC motor
3. Air gap – Space between stator and rotor where energy is transferred via induction.

2. AC MOTORS

Classifications

1. According to their principle of operation

a. Synchronous motor

b. Asynchronous motor

❖ Induction motors

- Squirrel cage induction motor
- Wound-rotor induction motor

2. AC MOTORS

Classifications

❖ Commutator motor

- Series motor – single phase, universal
- Compensated motor – conductively, inductively
- Shunt motor – single, compensated
- Repulsion motor – straight, compensated
- Repulsion-start induction motor
- Repulsion induction motor

2. AC MOTORS

Classifications

2. According to the type of current

- a. single phase

- b. three-phase

3. According to their speed

- a. constant speed

- b. variable speed

- c. adjustable speed

2. AC MOTORS

Classifications

4. According to their structural features

- a. Open
- b. Enclosed
- c. Semi-enclosed
- d. ventilated
- e. Pipe – ventilated
- f. Riverted frame

2. AC MOTORS

Types of AC Motors

1. Synchronous Motors — Speed remains constant with increasing load.
2. Asynchronous Motors — Speed varies with increasing load.

CHARACTERISTIC OF AN INDUCTION MOTOR

Induction (Asynchronous) Motor:

- AC motor where rotor current is induced electromagnetically by the stator's magnetic field.
- No electrical connection to the rotor is needed.

CHARACTERISTIC OF AN INDUCTION MOTOR

❖ Advantages

- It has a simple, robust and almost unbreakable construction
- It requires minimum care and maintenance
- Its cost is low and is very reliable
- It has a sufficiently high efficiency
- It has a reasonably good power factor
- It is self starting

CHARACTERISTIC OF AN INDUCTION MOTOR

❖ Disadvantages

- Its speed cannot be varied without loss of efficiency
- Its speed decreases with the increase of load
- Its has somewhat an inferior starting torque

GENERAL TYPES OF INDUCTION MOTORS

- Squirrel cage induction motor

Its rotor is self-contained and resembles a squirrel cage. The rotor conductors are permanently short-circuited by end-rings.



GENERAL TYPES OF INDUCTION MOTORS

- Wound rotor or phase wound or rotor induction motor or slip-ring motors

Its rotor has insulated coils of wire and resembles a DC armature. The rotor conductors are connected to slip rings and shorted thru the brushes.



DISTINCTION BETWEEN SQUIRREL CAGE AND WOUND ROTOR INDUCTION MOTORS

Squirrel cage induction motor

Wound Rotor Induction Motor

Copper losses is low

Copper losses is high

Starting torque is low but running torque is good

Starting torque is higher than squirrel cage induction motor

Efficiency is high due to low resistance

Efficiency is low

Speed can be varied by changing poles

Speed can be varied very slightly by changing the extra rotor resistance

Speed is nearly constant but slightly falls with load

More speed falls with load

Starting current is high

Starting current is low

Applications:

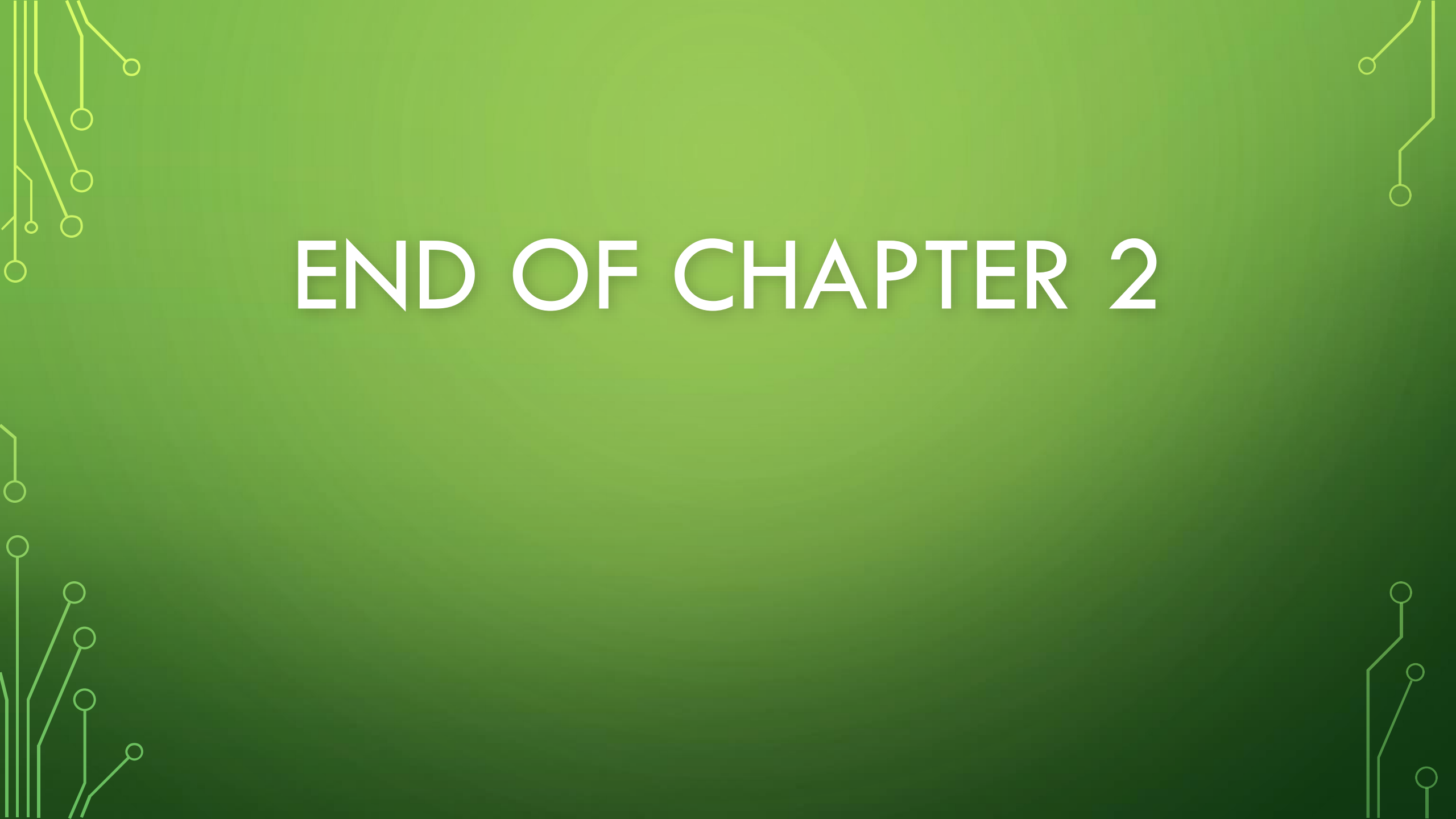
- Printing machine
- Flour mills
- Saw mills
- Lathe
- Blower mills

Applications:

- Rolling mills
- Flour mills
- Lifts
- Crane erector

REFERENCES

- Pertuzella, F. D. (2010). *Electric Motors and Control Systems* (1st ed.). New York: McGraw-Hill
- http://www.robotplatform.com/knowledge/actuators/dc_motors.html

The background is a solid green gradient. In the corners, there are decorative white line art elements resembling circuit traces or a stylized tree structure, with small circles at the end of the lines.

END OF CHAPTER 2